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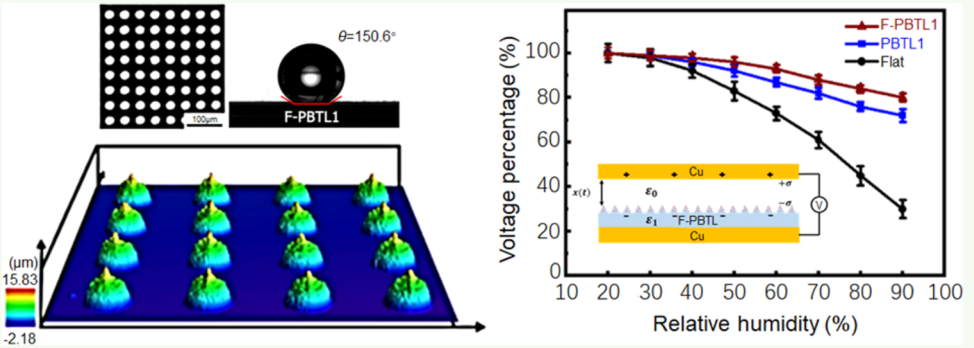
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Bioinspired Triboelectric Nanogenerator with High Humidity Resistance through Dual-Sized Morphology Construction

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ABSTRACT: Triboelectric nanogenerators (TENGs) exhibit significant potential for wireless sensor networks and portable power supplies due to their unique ability to convert environmental mechanical energy into electrical energy. Addressing the challenge of reduced TENG output performance in high-humidity environments, this study presents a lotus leaf-inspired poly(dimethylsiloxane) bionic triboelectric layer (PBTL) fabricated using poly(methyl methacrylate) (PMMA) molds processed by femtosecond laser and one-step templating. TENGs incorporating PBTLs exhibit significantly enhanced electrical output performance compared to those with flat surfaces by reducing the microstructural diameter, increasing the surface density, and raising the aspect ratio. Among the configurations tested, PBTL1 demonstrates optimal performance, achieving a short-circuit current of 2.15 μA, transferred charges of 15.16 nC, and an open-circuit voltage of 48.62 V—an improvement of 2.14 times over a flat layer. Furthermore, fluorination of the PBTL1 surface results in a superhydrophobic state with a water contact angle of 150.6°, maintaining 80% of its original output performance in a 90% high-humidity environment compared to only 30% for the flat layer. These findings suggest that the proposed methodology offers a straightforward and effective approach to designing high-performance and high-stability TENGs, leveraging the superior properties of micro-nanostructured PBTLs.

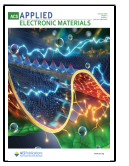
KEYWORDS: *triboelectric nanogenerator (TENG), micro-nanostructure, femtosecond laser, superhydrophobic surface, high-humidity environments*

1. INTRODUCTION

In the rapidly evolving information and communication technology landscape, the Internet of Things (IoT) has emerged as a pivotal direction for future global development. Addressing the challenge of energizing billions of IoT sensors, distributed energy harvesting—capable of extracting energy from the environment or through self-motion, presents a promising solution. Triboelectric nanogenerators (TENGs), leveraging the coupling effect of contact charge transfer and electrostatic induction,[1] are recognized as one of the most promising methods for self-powered small electronic devices due to their low cost, simple structure, and diverse energy sources. Currently, TENGs have been successfully applied to harvest energy from raindrops,[2,3] wind,[4,5] human activities,[6,7] and other mechanical sources.[8,9]

These miniature electronic devices, including sensors,[10,11] wearable devices,[12,13] and electronic skins,[14,15] operate in a variety of environments across numerous fields. Consequently, enhancing environmental stability and adaptability, particularly under harsh conditions, has become a critical focus for TENG development. Recent studies have highlighted the stability challenges of TENGs in humid environments. High humidity (above 80% RH) can lead to the formation of a water film at the friction interface, resulting in charge transfer or dissipation, thereby reducing electrical output performance and compromising stability.[16] These effects are closely linked to surfaces' hydrophilic and hydrophobic properties.[17] Superhydrophobic

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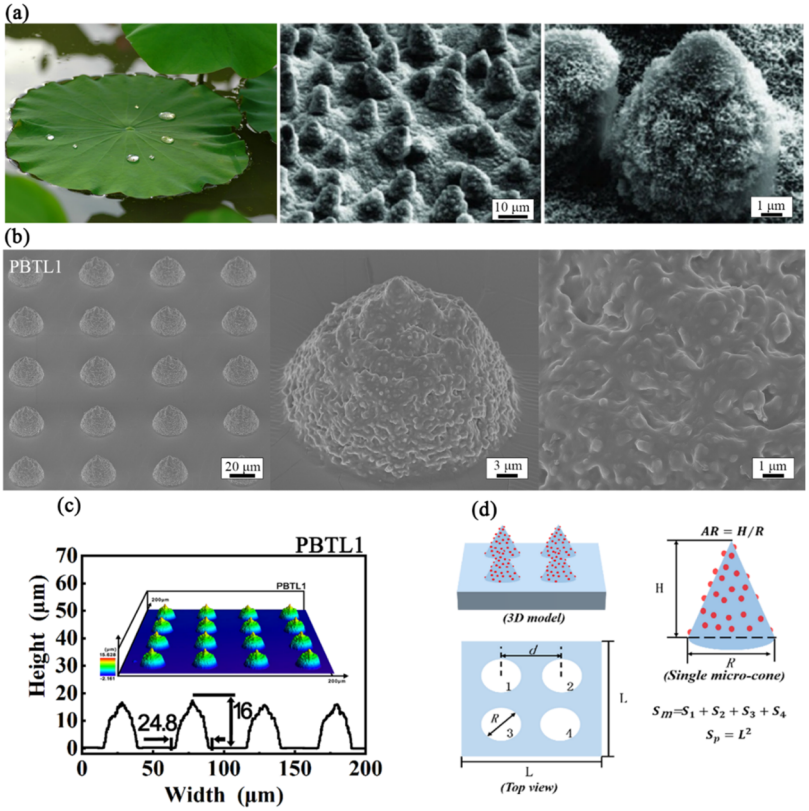


Figure 1. (a) SEM images of lotus leaf surface with micro-nanostructure in two different magnifications;[32] (b) SEM images of PBTL1 at various scales; (c) 3D surface morphology and cross-sectional profile of PBTL1; (d) simplified models of PBTL with micro-nanostructure, including a 3D model, top view of the micro-nanostructure, and a single geometry model.

surfaces, akin to lotus leaves, possess a low surface energy that prevents water layer formation, making them more suitable for humid environments. Inspired by the lotus leaf's superhydrophobic surface, the dual-sized morphology (micro-nanostructure) fabrication can significantly enhance surface hydrophobicity and improve TENG stability in humid conditions.[18] This hierarchical interlayer also increases effective contact areas for charge transfer, thereby enhancing TENG output performance.[19]

Despite the potential for self-cleaning abilities through natural material replication, achieving reproducibility and large-scale fabrication remains a significant challenge.[20] To improve TENG performance, researchers have explored various methods for artificial large-scale fabrications of micro-nanostructures on PDMS, including multistage photolithography,[21] multiple wet etching and electrospinning,[22,23] and force-assembling combined with electrostatic adsorption.[24] For instance, a PDMS template with a lotus leaf-like microstructure was fabricated by spin-coating a PDMS solution and curing it at high temperature, followed by replication using a copper film as a sacrificial layer, resulting in PDMS films with micro/nano dual-scale structures.[25] Another approach involved repeated etching of a silicon substrate followed by fluorination treatment to create hydrophobic micro/nanostructured surfaces. When combined with PDMS and Al electrodes, the electrical output performance of the TENG was significantly enhanced.[26] However, these methods typically require complex, time-consuming, and demanding multistep processes, limiting their widespread practical application.[27]

Laser processing has emerged as a viable surface processing technique, offering high precision, efficiency, and minimal environmental impact. Currently, femtosecond lasers[28,29] and CO_2 lasers[30,31] are employed to create polymer surface morphology in TENGs. Femtosecond laser direct writing on PDMS layers results in a microbowl structure, leading to higher costs and lower reproducibility of surface morphology across numerous samples.[32] CO_2 lasers are used to fabricate templates for replication to prepare triboelectric layers,[33] but the high power of CO_2 lasers typically results in millimeter-scale morphology and severe template ablation or destruction.

In this study, we introduce a high-precision, easy-controllable, and straightforward single-step process for fabricating stable dual-sized (micro-nanostructures) triboelectric layers, which significantly enhance the electrical output and humidity resistance of TENGs. Composite structured PBTLs were fabricated by replicating PMMA molds processed with femtosecond lasers. The lotus leaf-inspired TENG, assembled with these PBTLs, demonstrated a 2.14-fold increase in open-circuit voltage and a 5.1-fold increase in power output compared to flat TENGs, capable of simultaneously illuminating 63 commercial green LEDs. The enhanced hydrophobicity of the lotus leaf-inspired TENG maintained 72% of its original output performance in a 90% high-humidity environment. Furthermore, when the hydrophobic PBTL was modified by 1H, 1H, 2H, and 2H-perfluorodecyltriethoxysilane (PFDS), the fluorinated PBTL exhibited superhydrophilicity with a water contact angle of 150.6°, retaining 80% of its original output performance under 90% high humidity, whereas the flat TENG retained only 30%. Additionally, the lotus leaf-inspired TENG exhibited stable output performance over 10,000 cycles, ensuring exceptional durability.

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2. RESULTS AND DISCUSSION

2.1. Surface Morphology, Simplified Model Diagram, and Wettability. Figure 1a presents the SEM image of a natural lotus leaf at varying magnifications.[32] Figure 1b,c illustrates the SEM image and the 3D morphology of the PBTL1 surface, respectively. The surface of PBTL1 features conical protrusions with an approximate diameter of 24.8 μm and a height of 16 μm, covered with nanoscale particles ranging from 200 to 600 nm. This structure closely resembles the surface morphology of a lotus leaf. To elucidate the morphological parameters of the micro-nanostructure and examine their impact on the electrical output performance of TENG, Figure 1d provides a simplified model of PBTL surface morphology. This model includes a schematic 3D representation and a top view of the micro-nanostructure, highlighting a single microcone geometry model. In this model, R denotes the bottom diameter, d represents the spacing between the bottom centers of adjacent structures, L is the total length of the PDMS plane, and H signifies the height of the micro-nanostructure. The bottom area of micro-nanostructure on PDMS is defined as follows:

$$S_m = \pi\left(\frac{R}{2}\right)^2\left(\frac{L}{d}\right)^2 \quad (1)$$

The total area of the PDMS plane was calculated as follows:

$$S_p = L^2 \quad (2)$$

Consequently, the area occupied by the micro-nanostructures on the PDMS surface is

$$S_o = S_m/S_p = \pi\left(\frac{R}{2}\right)^2\left(\frac{L}{d}\right)^2/L^2 = \frac{\pi}{4}\left(\frac{R}{d}\right)^2 \quad (3)$$

According to eq 3, the morphology of the micro-nanostructure is determined by three factors: diameter R, density MD (defined as MD = R/d), and aspect ratio AR (defined as AR = H/R). Thus, the impact of morphological parameters on the electrical output performance is governed by these three factors. Comprehensive information about the fabricated samples (PTBL1−PBTL7) is summarized in Table 1.

Table 1. Structural Parameters of PBTLs

sample	average diameter R (μm)	average height H (μm)	average spacing d (μm)	micro-nanostructure density, MD = R/d	aspect ratio, AR = H/R
PBTL1	24.8	16.0	50.3	0.5	0.6
PBTL2	49.7	30.1	100.2	0.5	0.6
PBTL3	101.8	60.6	201.4	0.5	0.6
PBTL4	50.0	29.8	403.7	0.1	0.6
PBTL5	50.7	30.3	200.5	0.2	0.6
PBTL6	49.9	9.2	99.5	0.5	0.2
PBTL7	50.2	19.0	100.0	0.5	0.4

Figure 2 illustrates the static contact angles (SCAs) of water droplets placed on the flat and PBTL1−PBTL7, which are approximately 107, 138.1, 133.1, 120.6, 116.0, 125.0, 119.2, and 127.1°, respectively. These SCAs are all improved to varying degrees compared with the flat surface. This enhancement is attributed to the presence of surface micro-nanostructures, which trap a significant amount of air in the concave−convex voids, thereby reducing the contact area between water droplets and the surface, preventing droplet expansion, and enhancing surface hydrophobicity. Among them, PBTL1 exhibits the highest SCA (138.1°), which can be attributed to the maximum air entrapment in the micropores, leading to a reduced contact area between the surface and water droplets.[34,35] Considering that hydrophobic surfaces can effectively limit the formation of charge-dissipating moisture layers, these observations underscore the advantages of PBTLs in mitigating the decline in the electrical output performance of TENGs in humid environments.

2.2. Electrical Output Performance. The fabricated lotus leaf-inspired TENG in vertical contact-separation mode comprises Cu electrodes and PBTLs. Upon pressing, the upper Cu layer contacts the lower PBTL layer, resulting in the uniform distribution of equal and opposite charges on the two surfaces due to contact electrification. According to the triboelectric series, Cu becomes positively charged, while PBTLs acquire a negative charge. Electrostatic induction establishes a potential difference between the upper and lower electrodes when Cu is separated from PBTLs, driving charge transfer in the external circuit. When the separation distance reaches its maximum, an electrostatic equilibrium is achieved. Conversely, as Cu approaches PBTLs, the process reverses, causing charge flow in the opposite direction.

2.2.1. Influence of Microstructure Diameter R, Surface Density (MD), and Aspect Ratio (AR). Figure 3a shows optical microscope images and 3D morphologies under laser confocal microscopy of PBTL1, PBTL2, and PBTL3. All three surfaces exhibit similar conical structures, differing in size and distribution. The electrical output performance was measured by using a contact-separation test platform at a frequency of 1 Hz and an applied load of 10 N. Figure 3b−d illustrates that the electrical output performance of PBTLs with micro-nanostructures significantly surpasses that of the flat TENG. Among all electrode materials, the electrical output performance decreases with increasing base diameter R, with PBTL1 demonstrating the best performance. PBTL1 exhibits an open-circuit voltage (V_{oc}) of 48.62 V, a current of 2.15 μA, and a charge transfer (Q) amount is 15.16 nC, compared to the flat TENG's V_{oc} of 22.72 V, a current of 0.93 μA, and Q amount of 5.21 nC. Specifically, the V_{oc} increases by 2.13 times, the current by 2.31 times, and the charge transfer by 2.91 times. The primary reason for the enhanced electrical output performance of micro-nanostructured PBTL is the increased effective contact area between frictional layers, generating more frictional charges.

The operation of the triboelectric nanogenerator (TENG) is mainly divided into two processes: the contact process and the electrostatic induction process. Charge transfer occurs in the contact region; therefore, the simulation of the TENG can be classified into two main parts: contact mechanics simulation and electrostatic induction simulation. Based on the structure of the bionic TENG, a contact-separation bionic TENG model, as shown in Figure 4, is constructed. This model consists of a metal electrode (Cu) and a polymer (PDMS). The microstructure design adopts a conical shape, which results from the Gaussian distribution of the laser beam during fabrication, leading to microstructures with a Gaussian shape. However, due to the complexity of the Gaussian shape in simulation modeling, a conical shape, which closely resembles the Gaussian shape, is chosen as a simplified model. Additionally, nanoparticles are added to the surface of the

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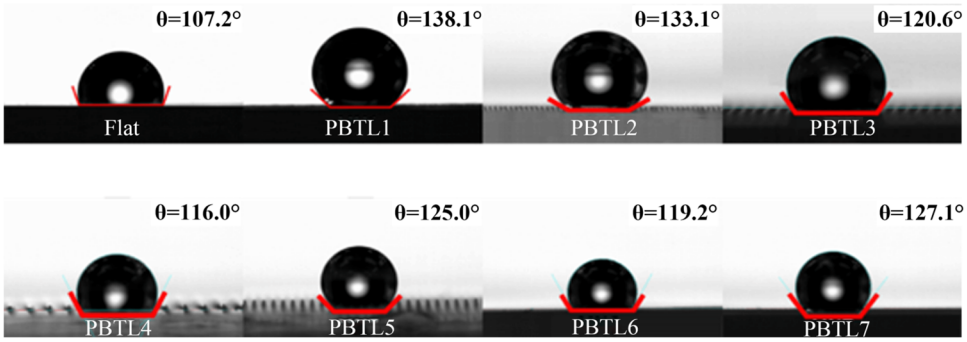


Figure 2. Water contact angles of the flat PDMS film and PBTL1−PBTL7.

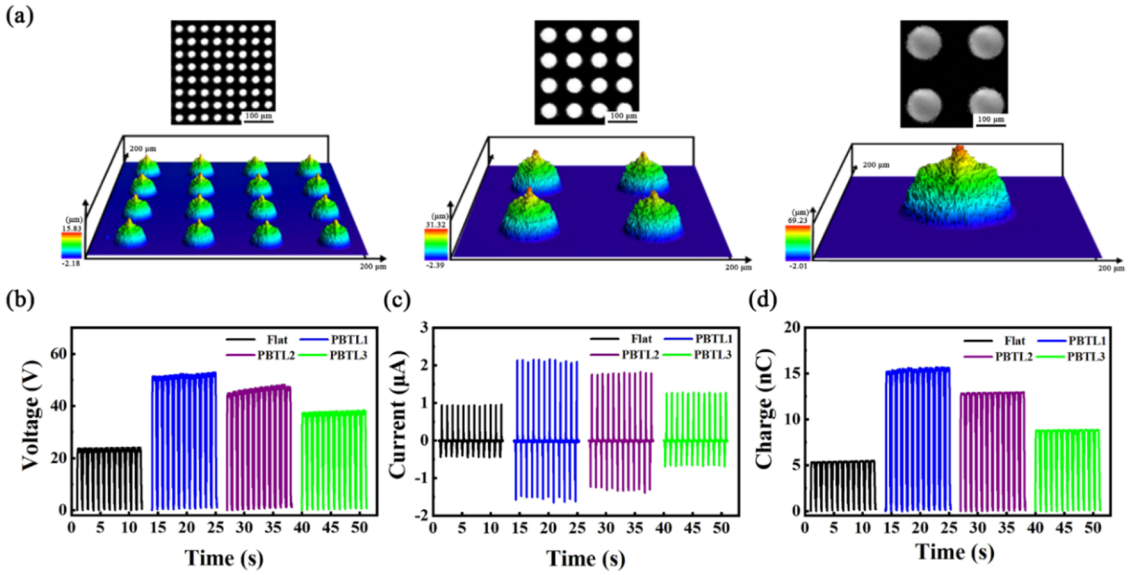


Figure 3. Output performance of TENGs. (a) Optical microscope and laser confocal microscope images of the PMMA templates. (b−d) Waveforms of open-circuit voltage, short-circuit current, and charge of TENG for the flat and PBTL1−PBTL3 configurations.

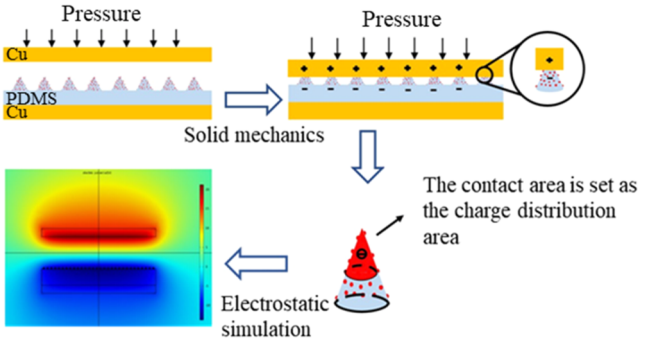


Figure 4. Schematic diagram of the simulation process.

microstructure. The simulation begins with the solid mechanics module to perform a steady-state contact mechanics simulation of the TENG. This step simulates the mechanical behavior during material contact, including factors such as contact pressure, contact area, and friction, to determine the charge transfer region. Next, the electrostatic induction process of the TENG is simulated, with the Cu electrode surface carrying a positive charge and the PDMS surface carrying an equal negative charge. To simplify the calculation, it is assumed that the charge is uniformly distributed within the contact deformation region and the charge attenuation effect is neglected.

To elucidate the contact interface behavior of PBTL1−PBTL3 under an applied load, mechanical deformation simulations were conducted using COMSOL. Figure 5a provides a schematic diagram of a 3D contact mechanics simulation for PBTL1−PBTL3. Figure 5b illustrates the distribution of contact pressure on the microstructured surface, with regions of nonzero contact pressure indicating contact areas. The overall contact area hierarchy is PBTL1 > PBTL2 > PBTL3. Figure 5c depicts the significant deformation of PBTL1−PBTL3 under the applied loads. The smaller diameter of PBTL1 facilitates pronounced deformation under external forces, ensuring full contact. In contrast, the deformations in the larger diameter PBTL2 and PBTL3 are comparatively less pronounced, resulting in only partial contact. Based on the calculated depression distance from COMSOL simulations, Figure 5d presents the effective contact area values of the flat and PBTL1−PBTL3, calculated based on the depression distance obtained from COMSOL simulations. Figure 5e compares the voltage peaks obtained from simulation and experimental results, demonstrating consistency between the two (Simulation parameters: dimension: 400 μm × 400 μm; thickness: 50 μm; uniform load 100 kPa; Cu is modeled as a rigid material with Young's modulus E_1 = 120 GPa and

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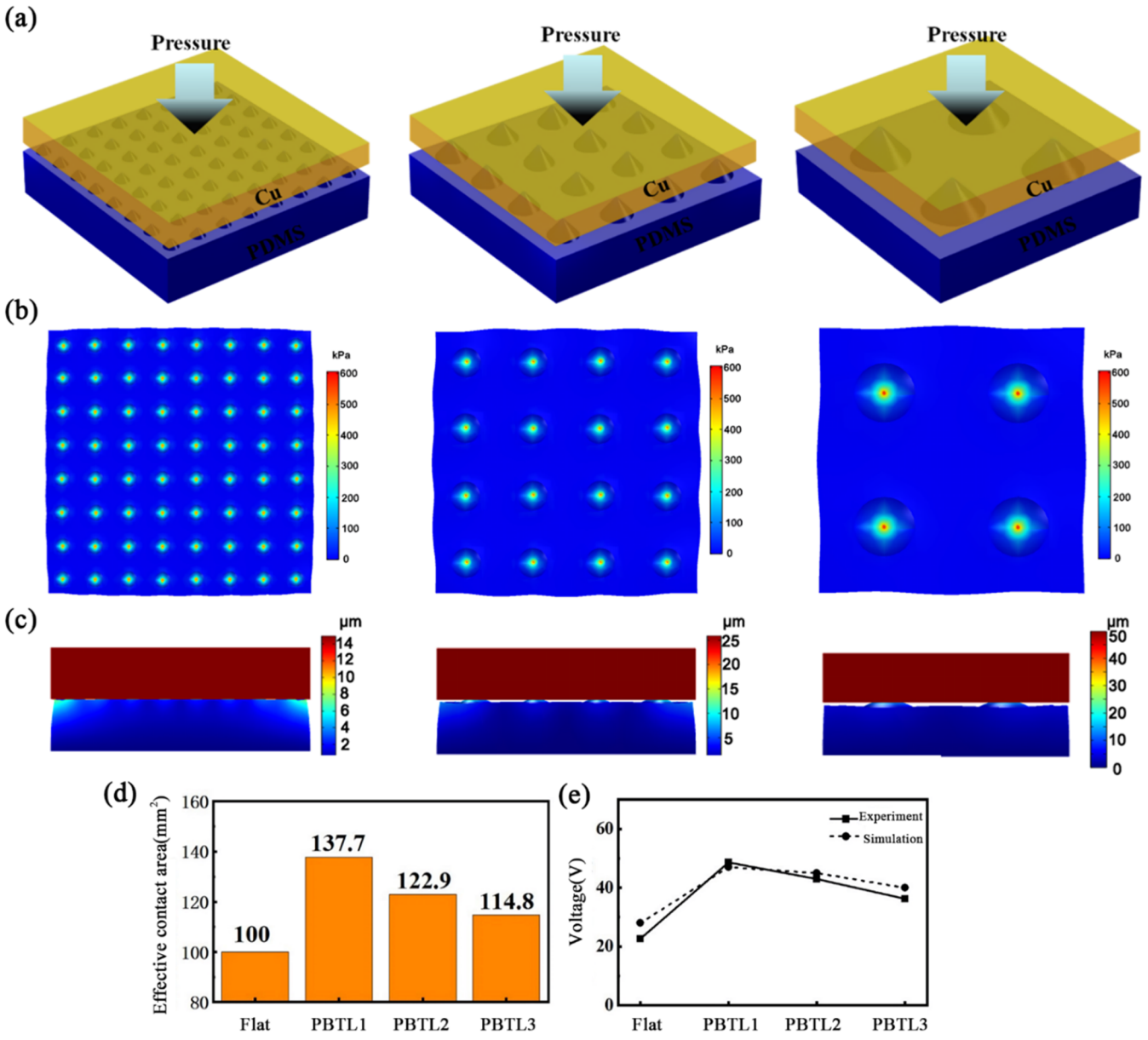


Figure 5. (a) Schematic mechanical simulation of COMSOL for PBTL1−PBTL3; (b) contact pressure distribution; (c) downward displacement plots obtained from simulation; (d) effective contact area values of the flat and PBTL1−PBTL3 TENGs; (e) comparison of simulation and experimental electrical performance.

Poisson's ratio ν_1 = 0.34; PDMS is modeled as a hyperelastic material described by a Mooney−Rivlin two parameter mode,[36,37] C_{10} = 291 kPa, C_{01} = 232 kPa; a uniform charge density of ±5 μC/m^2 is applied).

Figure 6a shows the optical microscope images and 3D morphologies under laser confocal microscopy of PBTL4, PBTL5, and PBTL2. The diameter of the circular hole array on the template remains constant, while the spacing between the hole centers varies. The electrical output performance results are depicted in Figure 6b−d. The V_{oc} values of Flat, PBTL4, PBTL5, and PBTL2 bionic PDMS films are 22.72, 27.62, 34.06, and 43.01 V, respectively. The corresponding short-circuit currents are 0.93, 1.14, 1.37, and 1.77 μA, and the charge transfer amounts are 5.21, 6.72, 8.92, and 13.08 nC. PBTL2 exhibits the highest output performance, with open-circuit voltage, short-circuit current, and charge transfer amounts being 1.9, 2, and 2.5 times those of the flat PDMS film, respectively. The enhancement in the electrical output performance of TENGs is attributed to the increase in surface density. As shown in the specific simulation and contact situation of MD, the higher electrical output is due to the increased effective contact area resulting from higher MD.

The influence of the aspect ratio (AR) on the electrical output performance of TENGs is investigated by testing PBTL6 and PBTL7 under identical conditions. Figure 7a presents the optical microscope images and 3D morphologies of PBTLs under laser confocal microscopy. The results of the electrical output performance are shown in Figure 7b−d. The V_{oc} values for the flat, PBTL6, PBTL7, and PBTL2 bionic PDMS films are 22.72 33.42, 38.24, and 43.01 V, respectively. The corresponding short-circuit currents are 0.93, 1.31, 1.56, and 1.77 μA, and the charge transfer amounts are 5.21, 8.48, 11.06, and 13.08 nC. PBTL2 exhibited the highest output performance in this test, and the electrical output performance

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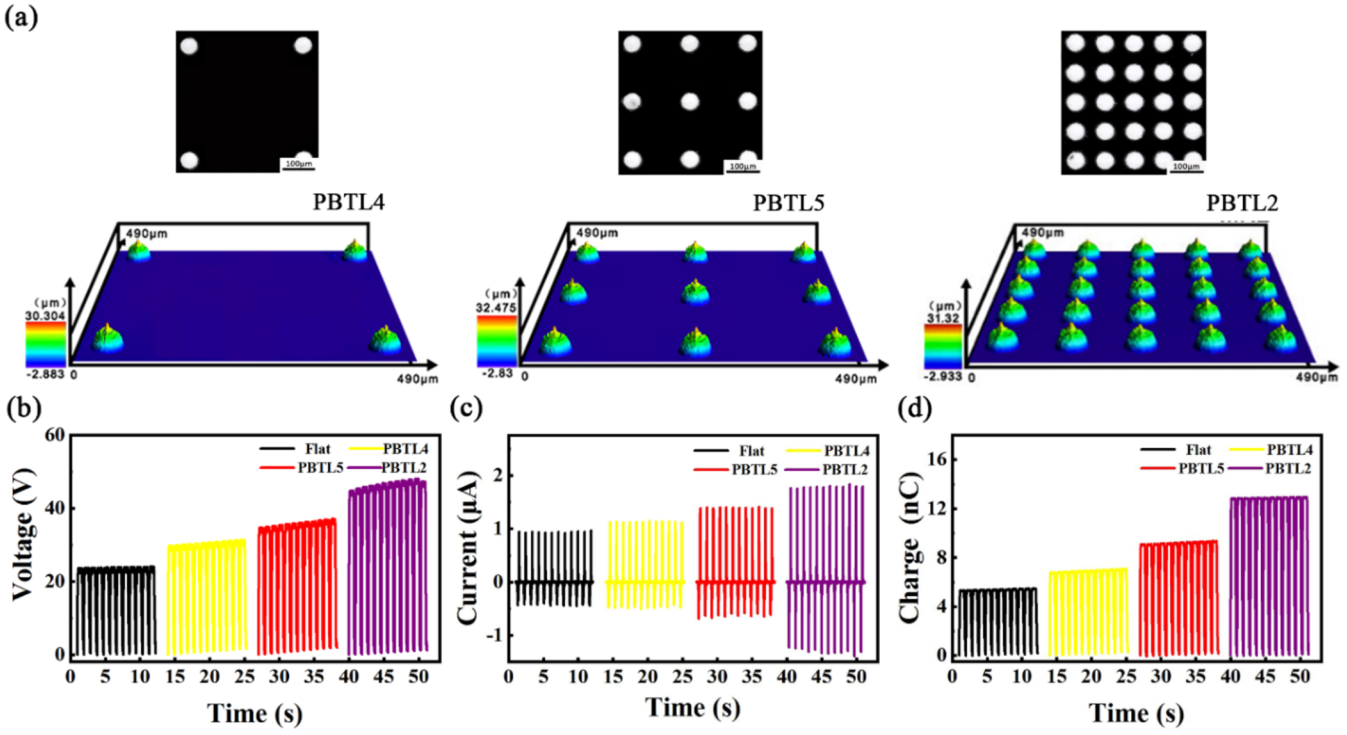


Figure 6. Output performance of TENGs. (a) Optical microscope and laser confocal microscope images of the PMMA templates; (b–d) waveforms of open-circuit voltage, short-circuit current, and charge of TENGs.

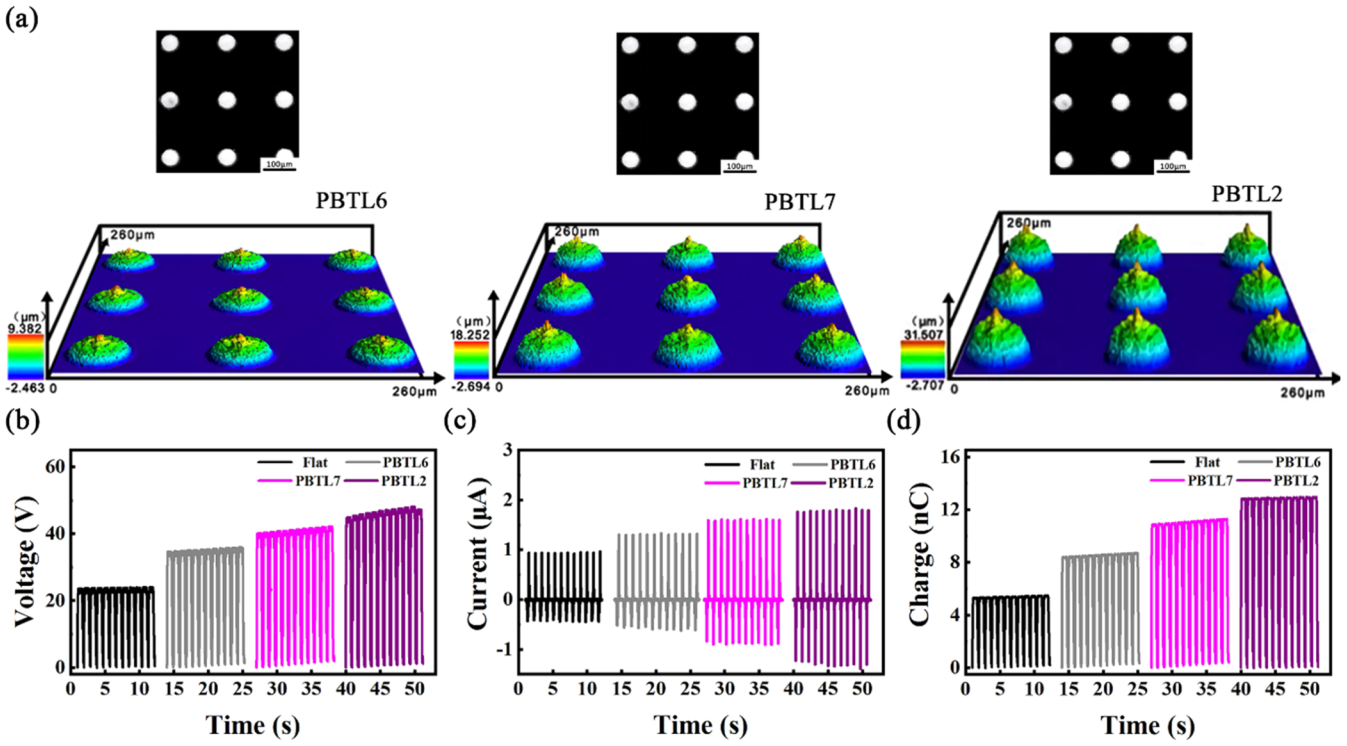


Figure 7. Output performance of TENGs. (a) Optical microscope and laser confocal microscope images of the PMMA templates; (b–d) waveforms of open-circuit voltage, short-circuit current, and charge of TENGs.

of TENGs improved with an increasing AR. From the specific simulation and contact situation of AR in Figure S1, it is evident that the increase in the aspect ratio enhances the contact area of individual microstructures, leading to an overall increase in contact area. This enhancement contributes to the improved electrical output performance of the bionic PDMS membrane.

2.2.2. Electrical Output Performance of PBTL1. The Flat PDMS and PBTL1 were assembled into flat TENG and lotus leaf-inspired TENG, respectively, as negative friction layers. Figure 8a shows the open-circuit voltage and short-circuit

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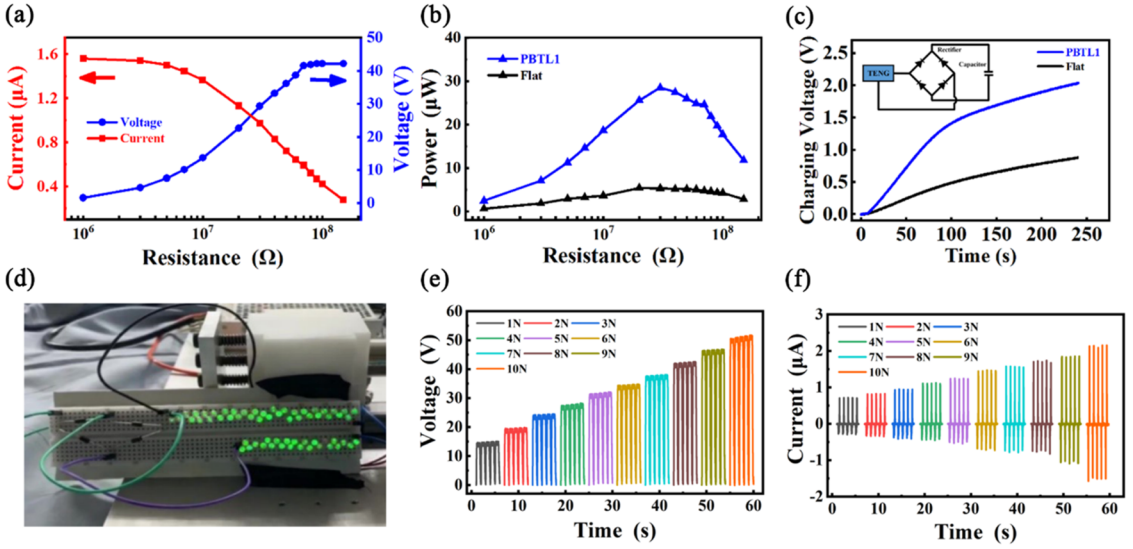


Figure 8. (a) Output voltage and current of TENG with PBTL1 under different external resistances. (b) Output power of PBTL1 and flat TENG under different external resistances. (c) Charging curves for a capacitance of 4.7 μF by PBTL1 and flat TENG. (d) Diagram of a full bridge rectifier for LEDs and digital photo showing 63 green LEDs lit. (e, f) Open-circuit voltage and short-circuit current of TENG with PBTL1 under different applied loads in the range of 1−10 N.

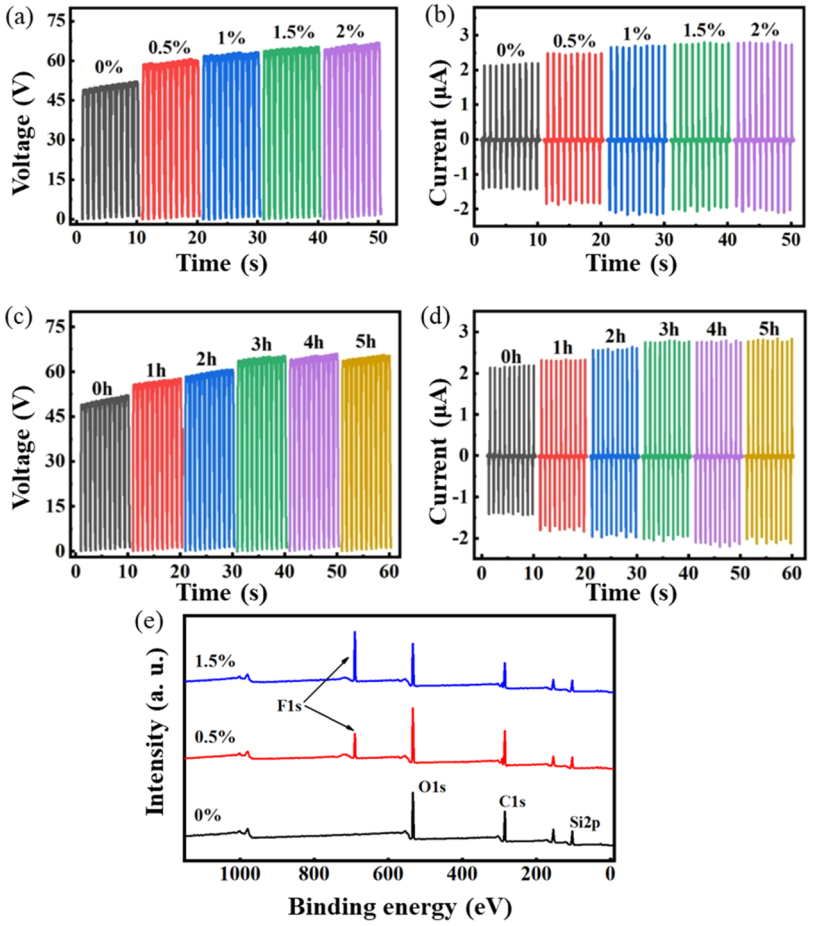


Figure 9. (a) Effect of fluoride solution concentration on the open-circuit voltage of bionic PDMS films. (b) Effect of fluoride solution concentration on the short-circuit current of bionic PDMS films. (c) Effect of fluoride solution fluorination time on the open-circuit voltage of PBTL1. (d) Effect of fluoride solution fluorination time on the short-circuit current of PBTL1. (e) XPS spectra of nonfluorinated and fluorinated bionic PDMS films.

current of the TENG under different loads when the contact pressure is 10 N, the separation distance is 2 mm, and the frequency is 1 Hz. The peak voltage increases with the gradual increase of resistance, while the peak current decreases due to

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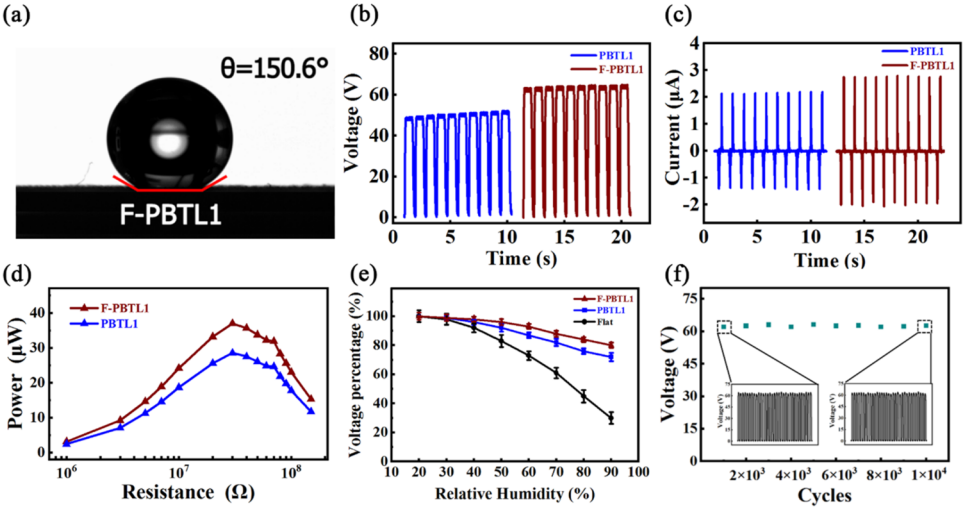


Figure 10. (a) Water contact angle of F-PBTL1. (b–d) Open-circuit voltage, short-circuit current, and output power of PBTL1 and F-PBTL1. (e) Voltage percentage of flat, PBTL1, and F-PBTL1 under different relative humidity conditions. (f) Robustness of F-PBTL1.

Ohm's loss. The instantaneous power, calculated in Figure 8b, rises in the low-resistance region and then falls in the high-resistance region. Consequently, the output power of PBTL1 reached a maximum value of 28.5 μW at an external load of 30 MΩ, which is 5.1 times higher than that of the flat TENG (5.5 μW). The flat TENG and lotus leaf-inspired TENG were used to charge a 4.7 μF capacitor at a frequency of 1 Hz. Figure 8c shows that the voltage of PBTL1 after 4 min of charging is 2 V, while the flat TENG's charging voltage is 0.8 V. The alternating current (AC) generated from TENG is converted to direct current (DC) by a full bridge rectifier to drive the LEDs, and the output of PBTL1 can simultaneously light up 63 commercial green LEDs (Figure 8d). Figure 8e,f demonstrates that the open-circuit voltage and short-circuit current exhibit a consistent increasing trend with stress, attributed to the promotion of contact area.[38]

2.2.3. Electrical Output Performance of Fluorinated PBTL. PBTL1, which exhibited the highest electrical output performance in the experiment, was selected for the fluorination treatment. The surface of the micro-nanostructured PBTL1 was treated with 1H,1H,2H,2H-perfluorodecyltriethoxysilane (PFDS). The testing conditions included a 10 N load, a separation distance of 2 mm, and a frequency of 1 Hz. Figure 9a,b presents the open-circuit voltage and short-circuit current results for PBTL1 films treated with different concentrations of fluorinated solutions. As the concentration of the fluorinated solution increases, the short-circuit current initially rises and then stabilizes. At a concentration of 1.5%, the open-circuit voltage and short-circuit current reach 62.5 V and 2.75 μA, respectively, representing 30 and 27% improvement compared to the untreated film. When the concentration increases from 1.5 to 2%, the electrical output performance stabilizes. Figure 9c,d shows the effect of different fluorination times with a 1.5% fluorinated solution on the electrical output performance of PBTL1. As the fluorination time increases, the short-circuit current initially increases and then stabilizes. When the fluorination time is less than 3 h, the output performance improves significantly with increasing fluorination time. After 3 h of fluorination, the open-circuit voltage and short-circuit current reach 62.5 V and 2.75 μA, respectively, representing improvements of 30 and 27% compared to the untreated bionic PDMS film MN1. For fluorination times greater than 3 h, the output performance shows no significant changes, as the fluorinated layer has already formed on the surface. This is because, when the fluorination time is short, the surface of the bionic PDMS film has not yet had sufficient time to form a dense fluorinated layer. As the fluorination time increases, more fluorine atoms attach to the surface. Once the fluorination time exceeds 3 h, the fluorinated layer stabilizes, leading to stable output performance. Figure 9e displays the XPS spectra of the bionic lotus leaf surface films treated with fluorinated solutions at concentrations of 0, 0.5, and 1.5%. The untreated bionic lotus leaf surface film shows three peaks at binding energies of 532.8, 284.91, and 102.5 eV, corresponding to O 1s, C 1s, and Si 2p, respectively. Following fluorination, an additional peak at 689.8 eV, representing F 1s, emerges, confirming the successful introduction of fluorosilane onto the surface. Comparing the XPS spectra of films treated with different fluorinated solutions, the peak height of F 1s increases at a concentration of 1.5% compared to 0.5%, indicating that a higher fluorine concentration results in more fluorine atoms attaching to the film's surface.

The fluorinated PBTL1 (F-PBTL1) achieved a superhydrophobic state with a water contact angle of 150.6°, as shown in Figure 10a. Figure 10b,c demonstrates that the open-circuit voltage and short-circuit current reach 62.5 V and 2.75 μA, respectively. Additionally, the maximum output power increases to 36.96 μW, representing a 29% enhancement over the nonfluorinated PBTL1, as shown in Figure 10d. These improvements in electrical output performance are attributed to the introduction of electron-attracting fluorine functional groups.[39,40]

We hypothesize that TENGs with superhydrophobic surfaces can effectively mitigate the degradation of the electrical output performance in high-humidity environments. To test this hypothesis, the electrical output performance of F-PBTL1 and PBTL1 was evaluated under varying relative humidity conditions ranging from 20% to 90%. As shown in Figure 10e, the open-circuit voltages of different surfaces decrease to varying degrees with increasing relative humidity due to charge dissipation. F-PBTL1, with its superhydrophobic surface, maintains 80% of its original output performance, experiencing only a 20% loss, while PBTL1 maintains 72% in a 90% high-humidity environment. In contrast, the flat surface

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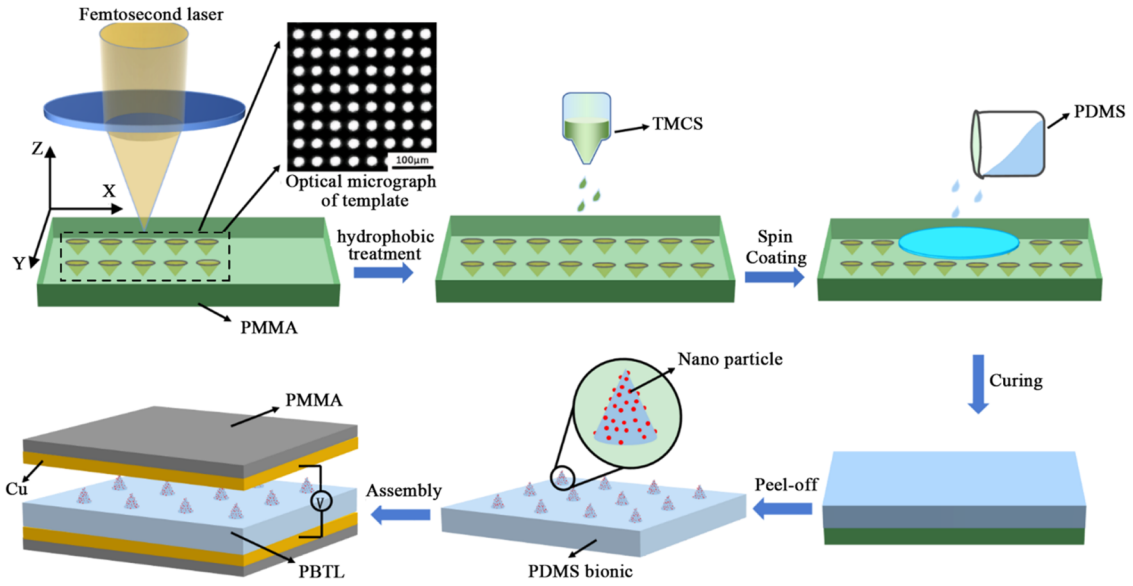


Figure 11. Schematic diagram for the fabrication processes of the lotus leaf-inspired TENG.

retains only 30% of its original output performance. Consequently, the superhydrophobic surface of F-PBTL1 effectively prevents charge dissipation and maintains output performance under high-humidity conditions. Finally, to assess robustness, the TENG with F-PBTL1 was subjected to continuous testing for 10,000 cycles, demonstrating stable output performance without significant degradation, as shown in Figure 10f.

3. CONCLUSIONS

In summary, dual-sized structures (micro- and nanostructures) analogous to the lotus leaf surface can be directly fabricated on the PDMS surface using femtosecond laser processing and one-step templating. The electrical output performance of TENG with PBTLs is markedly enhanced compared with TENGs with flat surfaces. The parameters of size, density, and aspect ratio of the micro-nanostructure influence the electrical output performance of TENG by creating varied contact situations. Notably, the TENG with PBTL1 exhibits optimal performance, achieving a 2.14 times higher voltage and a 5.1 times higher power than the flat TENG. Furthermore, the fluorinated surface attains a water contact angle of 150.6°, resulting in a superhydrophobic state that enables the lotus leaf-inspired TENG to maintain a high and stable output across diverse humidity conditions. Given the superior performance of these PBTLs with micro-nanostructures, we posit that this methodology offers a straightforward and effective approach to designing high-performance and high-stability TENGs.

4. MATERIALS AND METHODS

4.1. Materials. Poly(methyl methacrylate) (PMMA) plates were sourced from Nanjing Yishun Plastic Co., Ltd., Nanjing, China. Poly(dimethylsiloxane) (PDMS) was obtained from DOW CORNING Sylgard 184, comprising a prepolymer and a cross-linker, mixed at a weight ratio of 10:1. Trimethylchlorosilane (TMCS, purity 99%) and anhydrous ethanol were procured from Sinopharm Chemical Reagent Co., Ltd., Shanghai, China. 1H, 1H, 2H, and 2H-Perfluorodecyltriethoxysilane (PFDS) were purchased from Shanghai Chemical Co., Ltd., Shanghai, China. Acetic acid (CH_3COOH, 99.5%) was acquired from Jinan Xiaoshi Chemical Co., Ltd., Jinan, China.

4.2. Preparation of the PMMA Template. PMMA substrates were processed using an ultrafast laser micro-nanoprocessing center (MJ-Works, Moji-nano Technology Co., Ltd., China). This center utilized a regenerative amplified Yb femtosecond laser system with a pulse duration of 300 fs, a repetition frequency of 100 kHz, a central wavelength of 1030 nm, and an average power of 20 W. By adjustment of the fixed-point processing duration and scanning pitch, PMMA templates with varying morphological parameters were produced. The morphology parameters of the templates were measured using optical micrographs, as illustrated in Figure 11.

4.3. Fabrication of the PDMS Bionic Triboelectric Layer (PBTL). As shown in Figure 11, PMMA templates were cleaned with anhydrous ethanol using ultrasonic treatment and then dried. Subsequently, they were immersed in a TMCS culture dish for 30 min to achieve surface hydrophobic treatment. A PDMS mixture was then stirred thoroughly for 10 min and degassed in a vacuum for 30 min at room temperature to remove air bubbles. The PDMS mixture was spin-coated (400 rpm, 20 s) onto the PMMA templates and cured for 1 h at 90 °C to ensure complete solidification. The PBTL with micro-nanostructure was obtained by peeling it off from the PMMA molds. Seven different PBTLs (PBTL1−PBTL7) were fabricated based on the templates.

4.4. Fabrication of Fluorinated PBTL (F-PBTL). A uniform PFDS/ethanol solution was prepared by adding 0.15 g of PFDS and a few drops of 0.5% acetic acid to 9.85 g of ethanol solution and stirring for 30 min. The PBTLs were immersed in the PFDS/ethanol solution for 3 h after they were cleaned with deionized water and adequately dried. Finally, the films were flushed with distilled water and dried at 100 °C for 1 h.

4.5. Assembly of Lotus Leaf-Inspired TENG. PMMA was selected as the substrate, and copper foil with a thickness of approximately 100 μm served as both the top electrode and the positive triboelectric layer. PBTLs with micro-nanostructure were cut into regular sizes (10 mm × 10 mm × 140 μm) to act as the negative triboelectric layer. Another copper foil was placed at the bottom as an electrode to complete the assembly of the lotus leaf-inspired TENG, as schematically shown in Figure 11.

4.6. Characterization and Measurement. Optical micrographs of PMMA templates were captured using an optical microscope (AOSVI, Optics Instrument Co., Ltd., China). The surface morphology of PDMS was characterized by an SU8220 scanning electron microscope (SEM, Hitachi High-Tech Co., Ltd., Japan). 3D imaging of PDMS surface microstructure and contour dimensions

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measurements were performed by using a LEXT OLS5000 laser confocal microscope (Olympus, Japan). The contact angles were measured with an SDC-500 contact angle measuring instrument (SINDIN Precision Instrument Co., Ltd., China) to evaluate the film surface's wettability. The detection liquid was deionized water, and the droplet volume was 5 μL. The electrical performance of TENGs was assessed using an electrical property test platform in contact-separation mode, as shown in Figure S2. The open-circuit voltage and short-circuit current were measured using an electrostatic meter (Keithley 6514, Keithley Instruments).

■ ASSOCIATED CONTENT

Supporting Information

The Supporting Information is available free of charge at https://pubs.acs.org/doi/10.1021/acsaelm.4c02124.

Schematical mechanical simulation of COMSOL for PBTL6, PBTL7, and PBTL2; contact-separation TENG model; and the working mechanism of TENG (PDF)

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Notes

The authors declare no competing financial interest.

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